Financial Modeling

Class 3 — Portfolio Construction, Part II

Robust Approaches and the Treynor–Black Model: In-class session

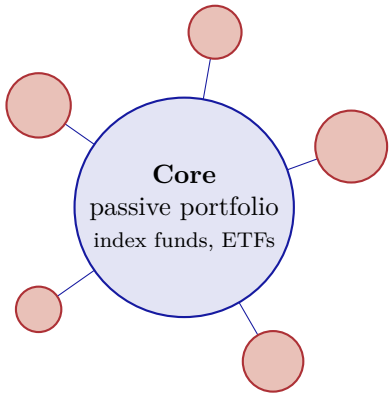
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Where we are, and today's two questions

- Last session: the optimizer maximizes the Sharpe Ratio given its inputs, and its weights are highly sensitive to small changes in them. This session completes the **construction** block
- **First half: allocating without forecasts.** The robust approaches (minimum variance, equal weight, value weight, risk parity) need no expected returns; forty years of data test them against the optimizer. Risk parity also **spreads the risk** across asset classes, which 60/40 does not
- **Second half: sizing positions on forecasts.** Treynor–Black states how far a forecast should move the portfolio from the passive allocation; the Fundamental Law ties performance to skill and breadth
- Common theme: knowing which parameters can be estimated reliably is part of the allocation decision

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Where Treynor–Black is used



Satellites: active positions, in securities or strategies with a forecast

- Many portfolios have a **core–satellite** structure: a core that holds the allocation passively, and smaller satellite positions wherever the manager has a forecast: securities, active strategies, external managers
- Treynor–Black sizes the satellites: each position, and all of them together
- Used for **security selection** and for sizing active strategies by managers with a tracking-error budget; not for the asset allocation

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Exercise: choose a construction approach

- The board has rejected the **unconstrained** optimizer, which shorts equities
- You still run the **CHF 2 billion** fund. Choose **one** construction approach for the long-run allocation:
  1. Mean-variance optimization within the mandate (no short positions)
  2. Minimum variance
  3. Equal weight
  4. Value weight
  5. Risk parity
- **In pairs, take three minutes to choose one approach and write down the reason for the choice**

Then vote on Wooclap. We vote again later in the class:

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The five approaches: weights and inputs

Strategy	Weights	Must estimate	Tangency portfolio if
Mean-variance	$x \propto V^{-1}\mu^e$	means, cov. matrix	inputs are exact
Minimum variance	$x \propto V^{-1}\iota$	cov. matrix	equal means
Equal weight	$x_i = 1/N$	nothing (rebalance)	equal means, vols, corr.
Value weight	$x_i \propto Cap_i$	nothing	CAPM holds
Risk parity	$x_i \propto 1/\sigma_i$	volatilities	equal SRs and corr.

$\propto$: proportional to; the weights are then scaled to sum to one. Cap_i: market capitalization of asset i.

- Each robust approach is the tangency portfolio **under an assumption of equality**: it treats as identical the parameters that are hard to estimate reliably
- Fewer estimated inputs $\Rightarrow$ fewer ways for estimation error to enter

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The four assets: a prediction

Four assets with a long return history, monthly data, 1978–2017:

	MSCI EAFE	S&P 500	US Credit	US Treasury
Volatility (annual)	17%	15%	7%	5%

Equal weight puts 50% of the *capital* in the two equity indices.

What share of the *risk* comes from them?

about 50% about 65% about 80%

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The four assets: three approaches with full-sample inputs

	MSCI EAFE	S&P 500	US Credit	US Treasury	Vol	SR
Equal weight	25%	25%	25%	25%	8.3%	0.42
risk contribution	**44%**	**38%**	12%	6%		
Risk parity	13%	15%	32%	41%	6.3%	0.51
risk contribution	23%	24%	29%	24%		
Mean-variance	0%	19%	4%	77%	5.4%	**0.55**

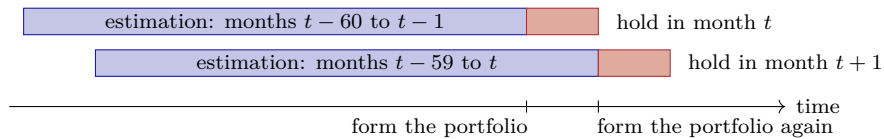
Inputs estimated once, on 1978–2017. Separate Excel file on iCorsi, sheet *In sample 1978-2017*.

- Equal capital is **not** equal risk: the two equity indices carry about 80% of the risk
- Mean-variance has the highest Sharpe Ratio **by construction**: this is **in-sample fitting**, the inputs and the evaluation use the same data. Does the ranking survive out of sample, when the inputs come from past data only?

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The horse race — the setup

- The same four assets: **MSCI EAFE, S&P 500, US Credit, US Treasury**, monthly data, 1978–2017
- Each month: estimate what needs estimating on the previous 60 months, form the portfolio, hold one month, repeat
- All results are out of sample, as an investor would have obtained them in real time



Before looking at the results, **we vote on Wooclap.**

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One vote before the results

Out of sample, 1978–2017: which of the five approaches had the highest Sharpe Ratio?

Mean-variance Min. variance Equal weight Value weight Risk parity

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Out-of-sample performance 1978–2017

	Raw return	Volatility	Sharpe Ratio
Risk parity	7.94%	6.19%	**0.53**
Minimum variance	7.06%	4.89%	0.49
Equally weighted	8.21%	8.32%	0.42
Value weighted	7.99%	11.55%	0.29
Mean variance	5.21%	14.73%	**0.04**

The computations are in a separate Excel file on iCorsi: *Comparison of portfolio formation approaches*, sheet *Portfolios*.

- Three of the five strategies outperform stand-alone US equity (SR 0.35): diversification adds value
- The strategy that requires the most estimation ranks **last**: mean-variance falls from 0.55 with full-sample inputs to 0.04 in real time

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Why does unrestricted mean-variance perform so badly?

- The tangency portfolio is a **non-linear** function of estimated means and covariances — estimation error is **amplified** by this non-linearity
- More assets ⇒ more parameters ⇒ more error. Here: 14 parameters from 60 monthly observations
- This is the result of the last session: *the weights react strongly to small errors in the inputs*. Here the effect is visible over 40 years of data
- The robust approaches perform better **because they require fewer estimates**

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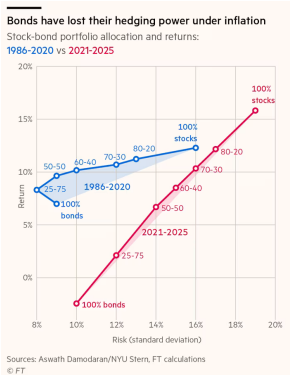
A caution: the average weights 1978–2017

	MSCI EAFE	S&P 500	US Credit	US Treasury
Value weighted	36%	41%	9%	14%
Equally weighted	25%	25%	25%	25%
Minimum variance	3%	8%	−47%	**136%**
Mean variance	0%	14%	27%	59%
Risk parity	12%	14%	33%	40%

- The strategies with fewer estimated inputs are all **overweight bonds** relative to value weights, and bonds benefited from a **40-year decline in interest rates**
- Risk parity overweights whatever has had low past volatility, that is, assets that did well recently and may be overvalued. In this sense it is **pro-cyclical**
- Bridgewater's All Weather fund: excellent 1996–2014 (8% return, 11% volatility), beaten by 60/40 afterwards

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The stock–bond correlation since 2021



Source: FT Unhedged, "Is the 60–40 portfolio dead?", August 2026; data A. Damodaran (NYU Stern).

- **The Fed put:** with low inflation, the Fed cut rates when equities fell. Bond prices rose, and bonds hedged equities
- **Blue curve, 1986–2020:** the frontier bends to the left, as the one built in Excel: 25% in stocks has less risk than bonds alone
- **Red curve, 2021–2025:** with inflation back, the Fed could not cut rates when equities fell. The stock–bond correlation turned **positive**, and the frontier is almost a straight line: **the diversification benefit disappears**
- This affects 60/40 *and* the horse-race winner: risk parity is **overweight bonds**

Back to your choice

At the start of the class you chose an approach.

Vote again: which approach now?

Mean-variance within the mandate Minimum variance Equal weight
Value weight Risk parity

We compare the two votes. If you changed your choice: which result of today made you change?

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Is there a right model?

- **No.** Each approach fails where its inputs fail: mean-variance on the estimated means and covariances; risk parity on past volatilities, so it overweights the recent winners
- Trust the inputs where they come from real work. Analyst research gives alphas that Treynor–Black can use: it is mean-variance optimization on an index model (next). A macro view, the subject of Class 7, tells you whether the recent winners will keep winning, before you lean on risk parity
- The safest allocation is a **weighted average of the allocations** from several approaches, with the weights set by your confidence in each one's inputs and in the outlook

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Treynor–Black: sizing the active positions

A different question. So far: how to divide the fund across asset classes. Now that portfolio is given (the passive portfolio, here the S&P 500). The analysts provide alphas for six stocks: how large should the **active deviations** be? (The stocks stand in for any active position with an alpha and a residual risk.)

	GE	MSFT	JNJ	K	BA	IBM
Alpha (per year)	+1.50%	−1.00%	−0.50%	+0.75%	+1.20%	+0.25%
Idiosyncratic variance	0.099	0.133	0.033	0.055	0.089	0.018

K's alpha is three times IBM's. By how much should K's position exceed IBM's?

about 3× about 2× about the same

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The answer: weights proportional to $\alpha_i/\sigma^2(\varepsilon_i)$

	GE	MSFT	JNJ	K	BA	IBM
α_i	+1.50%	−1.00%	−0.50%	+0.75%	+1.20%	+0.25%
$\alpha_i/\sigma^2(\varepsilon_i)$	+0.152	−0.075	−0.150	+0.137	+0.135	+0.136
Active weight	+45%	−22%	**−45%**	**+41%**	+40%	**+41%**

Active weight: weight in the active portfolio, the ratio divided by the sum of the six ratios (0.334).

- **IBM's position is equal to K's**, although its alpha is one third of K's: IBM's idiosyncratic variance is much smaller, so its alpha is a less noisy signal
- **JNJ's short position is twice MSFT's**, although its alpha is half of MSFT's, for the same reason
- The weight depends on the **alpha per unit of idiosyncratic variance**, not on the alpha alone

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Treynor–Black: MVO plus an index model

- **Step 1: an index model.** $R^e_{i,t} = \alpha_i + \beta_i R^e_{m,t} + \varepsilon_{i,t}$, with idiosyncratic shocks uncorrelated across stocks. Only $3N + 2$ parameters are needed: for the S&P 500 universe, **1,502** estimates instead of **125,750**
- It trades **model error** (industry co-movement is ignored) for less **estimation error**, as the robust approaches do
- **Step 2: the optimal portfolio** is the passive portfolio plus an **active portfolio** of the N bets. Its weight compares the alpha per unit of idiosyncratic variance with the market's reward-to-variance ratio:

$$w^0_A = \frac{\alpha_A/\sigma^2(\varepsilon_A)}{E(R^e_m)/\sigma^2_m} \qquad \text{then adjust for } \beta_A$$

- Within the active portfolio, each stock has a weight proportional to $\alpha_i/\sigma^2(\varepsilon_i)$. The sign of the weight is the sign of the alpha

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The Fundamental Law

- The squared Sharpe Ratio of the optimal portfolio is additive in the squared information ratios of the individual positions:

$$S_P^2 = S_m^2 + \underbrace{\sum_{i=1}^{N} \left[\frac{\alpha_i}{\sigma(\varepsilon_i)} \right]^2}_{IR^2_{\text{Active}}}$$

- Performance depends on two quantities: **skill** (the quality of the alphas) and **breadth** (the number of independent positions)
- If all alphas are zero, the passive portfolio is the optimal portfolio, which is the statement of the CAPM
- In the AMC, your Strategic Asset Allocation plays the role of the passive portfolio: the tactical positions add to its squared Sharpe Ratio only through their information ratios

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How many positions?

A manager targets an information ratio of 0.50.

Each position has an information ratio of 0.025.

How many independent positions are needed?

20 40 400

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Two ways to achieve the same Information Ratio

	Deep research	Systematic
Independent positions	4	400
IR per bet	0.25	0.025
Total IR ($\sqrt{N}\times$IR)	0.50	0.50
Typical example	Warren Buffett	quant funds (e.g. AQR)

- Same information ratio, different business models. At a given level of skill, doubling the information ratio requires **four times** as many independent positions
- The law applies at each level: asset allocation (timing more risk premia) and security selection (forecasting more alphas), as in the performance attribution of Class 2

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The full Treynor–Black portfolio: a prediction

We compute the Treynor–Black portfolio for the six stocks (market premium 6%, market volatility 19.5%, market Sharpe Ratio 0.306).

What share of the fund is invested in the active portfolio?

about 5% about 20% about 60%

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What the model says, with two sets of alphas

	S&P 500	GE	MSFT	JNJ	K	BA	IBM
Analyst alphas		1.50%	−1.00%	−0.50%	0.75%	1.20%	0.25%
weight	82.5%	8.0%	−3.9%	−7.9%	7.2%	7.1%	7.1%
Historical alphas		26.3%	25.1%	16.9%	3.2%	32.4%	16.0%
weight	−15.8%	13.7%	9.7%	26.0%	3.0%	18.7%	44.7%

The numbers come from the companion Excel file on iCorsi (Class 3 workbook, sheet *Treynor*).

- **Analyst alphas:** active portfolio 17.5% of the fund, Sharpe Ratio **0.317** against 0.306 for the market. The gain is small because the alphas are small. This order of magnitude is typical of active management
- **Historical alphas:** active portfolio **115%** of the fund, portfolio beta 0.08, Sharpe Ratio **2.17**, seven times the market's

Before implementing the second portfolio: which input is not credible, and why?

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Interpreting the result

- The model performs its task correctly. The problem lies in the inputs:
  - **Estimation error**: historical alphas mostly reflect noise. Using a past average as a forward-looking alpha is the same error as using the estimated equity premium in Part I, with larger consequences
  - **Tracking error**: the model maximizes the Sharpe Ratio and ignores the benchmark. With the historical alphas the beta is 0.08 and the **tracking error is 21%** a year against the S&P 500 (5.6% with the analyst alphas). In the AMC, rounds 2 and 3 have a tracking-error budget: the deviations must be sized to respect it
- The quality of the Treynor–Black portfolio depends entirely on the quality of the alphas: **garbage in, garbage out**
- The rest of the course deals with the inputs: the sources of defensible alphas, and the methods to test them rigorously

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Assignment 1: your allocation vs. the robust approaches

In round 1 of the AMC each team chooses its **Strategic Asset Allocation**. Add **half a page** to the round-1 report:

1. Compute the weights of your universe under **equal weight** and under **risk parity** ($\propto 1/\sigma_i$, volatilities from your own data)
2. Put the three sets of weights side by side — yours, equal weight, risk parity — each with its **risk contributions** and its **volatility**
3. One paragraph: where do the rules disagree with your weights, and what justifies your weights?

This section is graded as complete or incomplete, together with the round-1 report. The purpose is to identify *which positions depend entirely on an estimated expected return.*

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Assignment 2: state your alphas

In round 2 of the AMC each team deviates from its Strategic Asset Allocation with a few stocks, **long-only**, within the tracking-error budget of the guidelines. Add **one page** to the round-2 report:

1. **The stocks** — three to six. For each stock, an **alpha with its justification**: a fundamental valuation and the **catalyst** expected to realize it before the end of the challenge. A historical average is not an alpha
2. Weights among the stocks $\propto \alpha_i/\sigma^2(\varepsilon_i)$, idiosyncratic variances from your own data. **Long-only**: a negative alpha implies a weight of zero, not a short position
3. The expected **information ratio** from the Fundamental Law, what it adds to the Sharpe Ratio of your Strategic Asset Allocation, and a short assessment of its credibility

This page is graded as complete or incomplete, together with the round-2 report. In the final presentation, we will compare the realized contribution of the stocks with this page.

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Class 3 in summary

- **Part I:** the optimizer maximizes the Sharpe Ratio given its inputs, and amplifies any error in them
- **Part II, robust approaches:** in the absence of reliable forecasts, methods that estimate fewer parameters perform better out of sample
- **Part II, Treynor–Black:** given the passive portfolio, the active deviations are sized by the alpha per unit of idiosyncratic variance. The Fundamental Law gives the gain: the squared Sharpe Ratio rises by the sum of the squared information ratios
- **Next:** a practice session on the Strategic Asset Allocation and on the guidelines for the first round of the AMC

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